Lecture #12: Introduction to Bayesian Statistics

MATH 321 – Mathematical Statistics I

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November 9, 2026

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Bayes' Theorem

Recall Bayes' Theorem: if $A_1, \ldots, A_n$ form a partition of the sample space and $P(B) > 0$, then

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{P(B)} = \frac{P(B \mid A_i)P(A_i)}{\sum_{j=1}^{n} P(B \mid A_j)P(A_j)}.$$

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Bayes' Theorem

If X and Y are discrete random variables, we can easily express conditional probabilities and Bayes' Theorem using them. For any x with $P(X = x) > 0$,

$$\begin{aligned} P(Y = y \mid X = x) &= \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)} \\ &= \frac{P(X = x \mid Y = y)P(Y = y)}{\sum_t P(X = x \mid Y = t)P(Y = t)}. \end{aligned}$$

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Bayes' Theorem

But what if X and Y are continuous?

We can use the conditional density as our analogue for the conditional probability. For any y with $f_Y(y) > 0$,

$$\begin{aligned} f_{X|Y}(x \mid y) &= \frac{f_{Y|X}(y \mid x) f_X(x)}{f_Y(y)} \\ &= \frac{f_{Y|X}(y \mid x) f_X(x)}{\displaystyle\int_{-\infty}^{\infty} f_{Y|X}(y \mid t) f_X(t)\, dt}. \end{aligned}$$

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Rethinking Probability

Understanding how Bayesian statistics work requires a bit of a shift in how we think about probability, philosophically.

In the standard "frequentist" worldview, probability is the long-run relative frequency of an event. If you run an experiment many times, the proportion of times the event occurs out of the total number of tries will approach the "true" probability.

In the Bayesian framework, probability is instead a degree of subjective belief. We start with a belief, and we update it as we see new data. The more strongly I believe that an event will occur, the higher the probability I assign to it.

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Rethinking Parameters

We also want to rethink how we think about models and their parameters.

In the frequentist framework, the parameters of a distribution or a model are fixed constants that describe it. They exist, but they are unknown, so we need to estimate them.

To the Bayesian, because we don't know the value of a parameter, we describe our uncertainty about it with a probability distribution. That is, the parameter is treated as a random variable.

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Prior Distribution

Because the parameter $\boldsymbol{\theta}$ is treated as a random variable, it has a distribution, $g(\boldsymbol{\theta})$.

This is called the **prior distribution**.

It represents the statistician's belief about how the parameter is distributed, before we see the data.

It might be an uninformative flat prior, or it might be more specific, fitted to the situation, based on prior research or just a hunch.

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Likelihood of the Data

Then the data $\boldsymbol{x}$ enter through the likelihood function for our model, $\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x})$.

This is the probability (or probability density) of observing the data $\boldsymbol{x}$ when the parameter is $\boldsymbol{\theta}$.

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Posterior Distribution

Using Bayes' Theorem, we can find the **posterior distribution**, $g(\boldsymbol{\theta} \mid \boldsymbol{x})$:

$$g(\boldsymbol{\theta} \mid \boldsymbol{x}) = \frac{\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x}) g(\boldsymbol{\theta})}{\displaystyle\int_{\Theta} \mathcal{L}(\boldsymbol{t} \mid \boldsymbol{x}) g(\boldsymbol{t})\, d\boldsymbol{t}}$$

$$g(\boldsymbol{\theta} \mid \boldsymbol{x}) \propto \mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x}) g(\boldsymbol{\theta}),$$

where Θ is the set of all possible values of the parameter.

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Posterior Distribution

The **posterior distribution** represents our belief in the distribution of the parameters after we incorporate the data.

Because the posterior is a distribution for the parameter $\boldsymbol{\theta}$, we can then use it as the prior for the next experiment.

If we choose a prior such that, given a certain likelihood function, the posterior belongs to the same family of distributions as the prior, we call that a **conjugate prior**.

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Subjectivity

Subjectivity is the most frequent objection to Bayesian methods, as the prior distribution influences the conclusions.

Two scientists may arrive at different conclusions from the same data *using the same statistical analysis*, if they start with different priors.

However, it can be shown that the influence of the prior goes to zero as the sample size increases (for all but the most bone-headed priors, such as a prior that gives zero probability to the true value of the parameter).

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Example

Suppose that, before the November 2025 election for Mayor of New York City, we wanted to estimate the proportion of the vote Zohran Mamdani would get. We ask $n = 100$ likely voters whether they are going to vote for Mamdani, assuming that their responses $x_1, \ldots, x_n$ are independent Bernoulli trials with probability θ of supporting Mamdani. 62 of those asked said they would vote for Mamdani.

So the likelihood is:

$$\begin{aligned}\mathcal{L}(\theta \mid \boldsymbol{x}) &= \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} \\ &= \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}.\end{aligned}$$

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Example

Let's be smart about our prior distribution.

The success probability θ can take on any value between 0 and 1, so we need a distribution with that as the support. Can you think of a good choice?

The Beta distribution should work perfectly!

$$g(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \leq \theta \leq 1.$$

Let's use a prior distribution of Beta($\alpha = 10, \beta = 10$).

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Example

We only need the part of the posterior density that involves the parameter, so we treat it as **"proportional to"** rather than equal to.

Because the posterior is a PDF, the missing constant is whatever makes the full density integrate to 1.

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Example

$$
\begin{aligned}
g(\theta \mid \boldsymbol{x}) &\propto \mathcal{L}(\theta \mid \boldsymbol{x}) g(\theta) \\
&= \left(\theta^{\sum x_i}(1-\theta)^{n-\sum x_i}\right)\left(\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}\right) \\
&\propto \theta^{\sum x_i+\alpha-1}(1-\theta)^{n-\sum x_i+\beta-1}
\end{aligned}
$$

Notice that if we set $\alpha^* = \sum x_i + \alpha$ and $\beta^* = n - \sum x_i + \beta$, then this is of the form $\theta^{\alpha^*-1}(1-\theta)^{\beta^*-1}$, which matches the Beta distribution!

Thus, the Beta prior is a **conjugate prior** for the Bernoulli (or binomial) likelihood.

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Example

Thus, the posterior for our example is Beta(α^*, β^*) with $\alpha^* = \sum x_i + \alpha = 62 + 10 = 72$ and $\beta^* = n - \sum x_i + \beta = 100 - 62 + 10 = 48$.

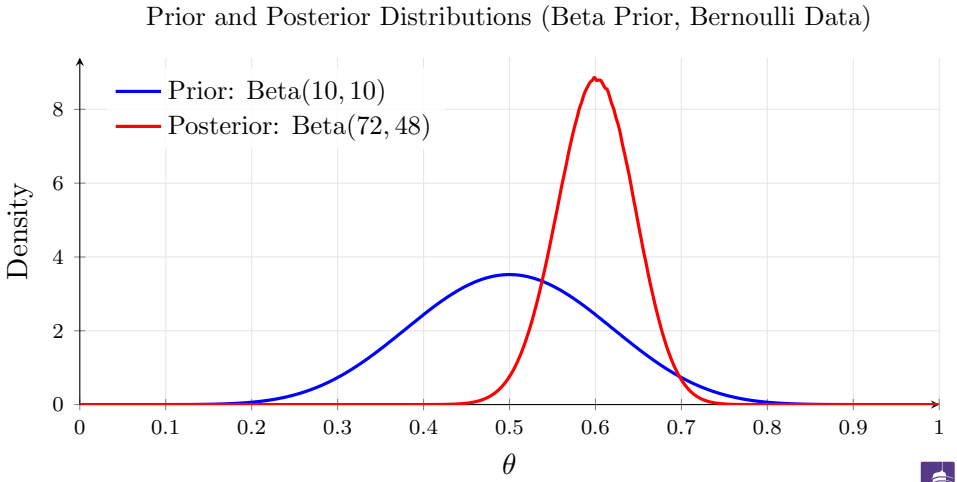
We can use this to estimate θ:

$$\widehat{\theta} = E[\theta \mid \boldsymbol{x}] = \frac{\alpha^*}{\alpha^* + \beta^*} = \frac{72}{72 + 48} = \frac{72}{120} = 0.60.$$

We can also use the distribution to get a **credible interval** (Bayesians don't make confidence intervals) or to perform hypothesis tests.

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Example



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Simulating the Posterior Distribution

If the posterior distribution is from a known family, simulating from it is pretty trivial: `R` and Python both have functions to draw random values from known distributions.

If the posterior is not from a known family, it can be really hard.

We use algorithms designed to make this easier, namely **Markov chain Monte Carlo (MCMC) samplers**.

Two of the most common are the **Metropolis–Hastings algorithm** and the **Gibbs sampler**.

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